

ECON 100B - SECTION 1

INTRODUCTION, SYLLABUS, AND BASIC CONCEPTS

Felix Zhang

AGENDA

- Introductions
- Section Logistics + Syllabus
- Why Study Macro?
- Key Concepts

ABOUT ME

Felix Zhang

- 4th year undergrad @ Berkeley. Economics, Data Science, and City Planning
- Previous experience: Research on economic case for fossil fuel divestment, ESG finance
- From Philadelphia, PA (Good riddance, Kevin Patullo)



INTRODUCTIONS

Find a partner, introduce yourselves and when we regroup, present your partner to the class.

- Name
- Major
- Hometown
- Favorite Restaurant in Berkeley

SECTION SYLLABUS

LOGISTICS

Discussion Sections

- Starts at Berkeley time
- Slides (generally) uploaded day before class
- Will review lecture material, work through problems including PSET and former exam questions
- Discuss current news and articles related to coursework

Office Hours

- 3-5PM on Mondays
- In-person in Evans 542 unless otherwise announced
- What should I come to office hours for?
- Questions about content, logistics, or Berkeley econ advice

OBJECTIVES

My goals for this discussion section

- Help you learn to solve problems and prepare for the exams
- Make macroeconomics interesting and relevant beyond the course material
- Give you a starting point to learn more about the key ideas and develop a more **mature macroeconomic intuition**

CONTACTING ME

Emails to felixzhang04@berkeley.edu are better than bCourses messages - I check email multiple times throughout the day

- Include the subject line ECON100B, which helps me search for your email if I lose it
- Give me 48 hours to respond and if I still haven't, don't hesitate to reach out again

WHAT COUNTS AS MACROECONOMICS?

Macro - comes from the Greek word *makros*, which means large. Thus "macro" economics is about the big stuff in the economy (the aggregate variables)

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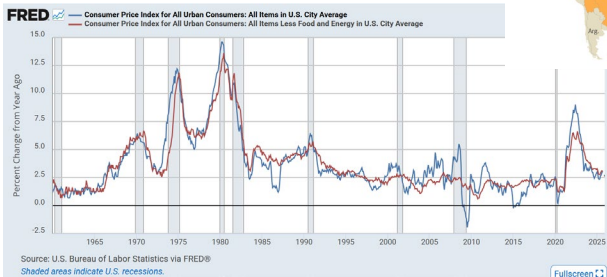
Macro - comes from the Greek word *makros*, which means large. Thus "macro" economics is about the big stuff in the economy (the aggregate variables)

- Recessions, interest rates, and inflation
- Long-term growth and technological innovation
- International trade
- Financial stability (ex. 2008 crisis)
- Labor markets and employment

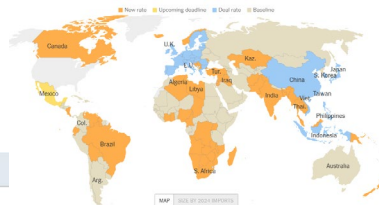
WHY STUDY MACROECONOMICS

Supreme Court Hearing Reveals Unease Over Threats to Fed Independence

As the justices weighed the consequences of allowing President Trump to fire a Federal Reserve official, the president reprised his pressure campaign on the central bank.



Here's the Latest Trump Tariffs on Countries and Products



WHY STUDY MACROECONOMICS

Macroeconomics affects us everyday and the language of macroeconomics is used throughout industry, politics, and the media

- Affects your job, your business, and your retirement plans
- Necessary for reading and understanding the news
- Implications for policy and people's wellbeing

KEY CONCEPTS

Introducing some words we'll use a lot in this course that you may have heard before but aren't quite clear on:

- Models
- Equilibrium
- Exogenous vs. Endogenous
- Growth vs. Business Cycles
- Closed vs. Open

MODELS

What is an economic model?

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- A thought experiment for when we can't run real experiments
- An argument with guaranteed validity (though not soundness)

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MODELS

A practical definition is that models are a **bunch of equations describing the rules of behavior** of different kinds of people (households, companies, governments etc.) in the economy. Solving those equations gives us some testable prediction about what the world will look like and what people's behavior depends on.

EXAMPLE

A supply and demand model for apples

- Demand curve: $P = -2Q_d + 5$ describes how many apples Q_d consumers want at price P
- Supply curve: $P = 3Q_s + 1$ describes how many apples Q_s farmers are willing to sell at P
- Solving $Q_d = Q_s$ gives us a prediction for the price and quantity

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PERSPECTIVES ON MODELS

"All models are wrong. Some are useful."

- George Box, statistician

EQUILIBRIUM

Equilibrium is “**what happens**” in the model

- In macro, it is essentially a **prediction** satisfying certain conditions

EQUILIBRIUM IN EXAMPLES

Equilibrium is “**what happens**” in the model

- In the apple example, the equilibrium price and quantity are the ones that satisfy the consumer's demand curve and firm's supply curve
- In other models, the EQ variables might be how much a person works, how much you should invest in the company etc.
- Ex. If (1) firms choose to hire to maximize profit, (2) households choose how much to work to maximize utility, then how many hours of work get done in the economy?

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EXOGENOUS VS. ENDOGENOUS

Exogenous

- "Outside" the model
- A parameter, ideally something you can measure
- A number you already know, that you use to explain something else

Endogenous

- "Inside" the model
- Something you don't know that you wish to explain using what you know

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- Price P and quantity demanded Q_d are endogenous, they are determined by the model
- The numbers -2 and 5 are given exogenously, we know them already

EXAMPLE

For a more concrete example:

$$\text{Crop_Output} = \alpha \text{Rainfall} + \beta \text{Fertilizer_Type}$$

In this model, which variables are endogenous, and which are exogenous?

GROWTH VS. BUSINESS CYCLES

Growth - describes the **long-run** trajectory of a country

- Why are some countries poor and others wealthy?
- How did China grow so fast and so quickly?
- Which countries will have the largest economies in 10 years?

GROWTH VS. BUSINESS CYCLES

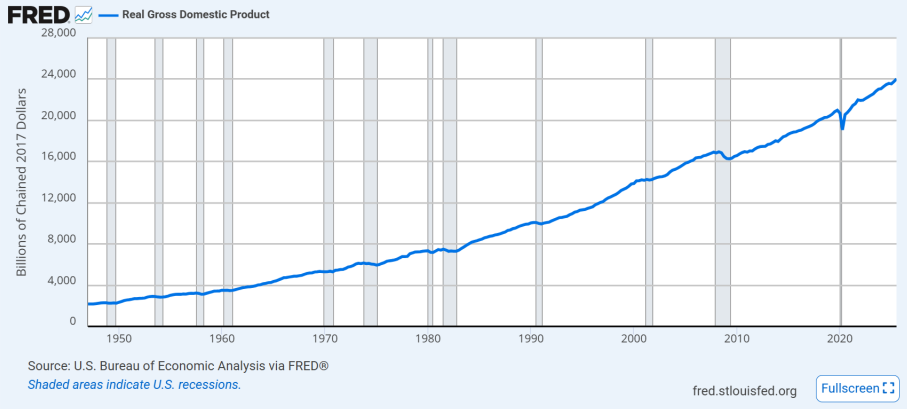
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Cycle - describes the **short-run** fluctuations, ie. recessions and booms

- Why did the Great Depression happen?
- Why do stock markets crash?
- How does the central bank affect the economy?

GROWTH VS. BUSINESS CYCLES



CLOSED VS. OPEN

Closed - one country, focused on impacts of domestic factors

- growth models (in this class)
- business cycles (first part of this class)

Open - two or many countries, focused on the relationship between them

- trade issues
- cross-country capital flows

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A LOOK AHEAD

Section #	Dates	Topics to Look At
1	1/22, 1/26	Introduction, Key Terms
2	1/27, 1/28	Math Review, The Long Run
3	1/29, 2/2	<u>The Production</u> Model Pt. 1
4	2/3, 2/4	<u>The Production</u> Model Pt. 2
5	2/5, 2/9	Solow Growth Model Pt. 1
6	2/10, 2/11	Solow Growth Model Pt. 2, PSET Practice